

MatMul — C = A × B on the T1 fabric (3×3 worked example)

Steps run top → bottom. Colour helps tracking here. The V-load transpose turns columns into B^T rows can be seen with colour; each Output cell keeps its final-position colour, so a result enters at column 0 and shifts right into place.

Example

0. Reference

C = A × B (true result)

Matrix A

1	2	3
4	5	6
7	8	9

Matrix B / B^T

B

10	11	12
13	14	15
16	17	18

Intermediate

Output C

84	90	96
201	216	231
318	342	366

Expected

Setup

1. Load H / V

vle8 v4,(A) [H]
csrwi 0x7c0,1
vle8 v5,(B) [V] → B^T

Matrix A

1	2	3
4	5	6
7	8	9

Matrix B / B^T

B^T

10	13	16
11	14	17
12	15	18

Intermediate

Output C

Iteration 0 (Full Steps)

2. k=2 Broadcast

csrwi 0x7c0,1
vrgather.vx v6,v5,2 [V]

Matrix A

1	2	3
4	5	6
7	8	9

Matrix B / B^T

10	13	16
11	14	17
12	15	18

→

12	15	18
12	15	18
12	15	18

Intermediate

Output C

3. k=2 Multiply

csrwi 0x7c0,0
vmul.vv v6,v4,v6 [H]

Matrix A

1	2	3
4	5	6
7	8	9

Matrix B / B^T

10	13	16
11	14	17
12	15	18

→

12	30	54
48	75	108
84	120	162

Intermediate

Output C

4. k=2 Reduce

vredsum.vs v7,v6,v28
→ v7[0]=C[*][2]

Matrix A

1	2	3
4	5	6
7	8	9

Matrix B / B^T

10	13	16
11	14	17
12	15	18

→

12	30	54
48	75	108
84	120	162

→

96		
231		
366		

Intermediate

Output C

5. Shift Right

vslideup.vi v8,v7,1
vmv.v.v v7,v8

Matrix A

1	2	3
4	5	6
7	8	9

Matrix B / B^T

10	13	16
11	14	17
12	15	18

→

Intermediate

Output C

->	96	
->	231	
->	366	

shift one column right →

SAME AS ABOVE

6. Broadcast, 7. k=1 Multiply

Iteration 1 (Simplified)

8. k=1 Reduce

vredsum.vs v7,v6,v28
→ v7[0]=C[*][1]

Matrix A

1	2	3
4	5	6
7	8	9

Matrix B / B^T

10	13	16
11	14	17
12	15	18

→

11	28	51
44	70	102
77	112	153

→

90	96	
216	231	
342	366	

Intermediate

Output C

9. Shift Right

vslideup.vi v8,v7,1
vmv.v.v v7,v8

Matrix A

1	2	3
4	5	6
7	8	9

Matrix B / B^T

10	13	16
11	14	17
12	15	18

→

Intermediate

Output C

->	90	96
->	216	231
->	342	366

shift one column right →

SAME AS ABOVE

10. Broadcast, 11. k=0 Multiply

Iteration 2 (Simp)

12. k=0 Reduce

vredsum.vs v7,v6,v28
→ full C
vse8 v7,(C)

Matrix A

1	2	3
4	5	6
7	8	9

Matrix B / B^T

10	13	16
11	14	17
12	15	18

→

10	26	48
40	65	96
70	104	144

→

84	90	96
201	216	231
318	342	366

Intermediate

Output C

Answer!!